United States Patent Application Publication	20250285069
Kind Code	A1
Publication Date	September 11, 2025
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UTILIZING BEHAVIORAL ELICITATION EVENTS FOR EVALUATING A TRAINING PERFORMANCE AGAINST A COMPETENCY FRAMEWORK

Abstract

One example provides a method of evaluating a training performance of a training subject and performed in a training environment. The method receives a simulation state update from a computing system of the training environment, and queries a behavioral elicitation event (BEE) database to determine a set of active BEEs based upon the simulation state update. The method further receives a plurality of simulation action notifications from the computing system, and for each simulation action notification, determines whether an action represented by the simulation action notification matches an elicited action for an active BEE. The method further stores information regarding the elicited action as a response to the active BEE. The method determines a set of completed BEEs comprising any active BEEs that have all corresponding elicited actions met, and for each completed BEE, stores information regarding the completed BEE.

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Family ID: **96949202**

Appl. No.: **19/212409**

Filed: **May 19, 2025**

Related U.S. Application Data

parent US continuation-in-part 18364380 20230802 PENDING child US 19212409

Publication Classification

Int. Cl.: **G06Q10/0639** (20230101); **G06F3/0482** (20130101); **G09B9/08** (20060101)

U.S. Cl.:

CPC **G06Q10/06398** (20130101); **G06F3/0482** (20130101); **G09B9/08** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 18/364,380, filed Aug. 2, 2023, the entirety of which is hereby incorporated herein by reference for all purposes.

FIELD
[0002] The disclosed examples relate to automated evaluation of a training subject against a competency framework, and user interfaces for the automated evaluation.
BACKGROUND
[0003] Some industries evaluate professionals against a specified competency framework during professional training. The competency framework can comprise soft skills to be demonstrated by the training subject in addition to hard skills performed during a training performance in a training environment. Example soft skills include communication, leadership, teamwork, situation awareness, management of information, management of hard skill tasks, and knowledge of hard skill procedures. For example, training of commercial pilots uses a competency-based training and assessment (CBTA) to evaluate student pilots on core pilot competencies defined by the International Civil Aviation Organization (ICAO). The ICAO currently evaluates trainee pilots based upon nine pilot competencies: application of knowledge, application of procedures and compliance with regulations, communication, airplane flight path management for automation, airplane flight path management for manual control, leadership and teamwork, problem solving and decision making, situation awareness and management of information, and workload management. Current methods of evaluating soft skills are performed by observation of an instructor during the training performance. Such evaluation by observation can be difficult to objectively assess and/or provide feedback to the training subject for the instructor.
SUMMARY
[0004] One disclosed example provides a method of evaluating a training performance of a training subject against a competency framework. The training performance is performed in a training environment that simulates a physical environment. The method comprises receiving a simulation state update from a computing system of the training environment, and querying a behavioral elicitation event (BEE) database to determine, from a plurality of BEEs stored in the BEE database, a set of active BEEs based upon the simulation state update. Each BEE of the plurality of BEEs comprises one or more elicited actions and one or more observable behaviors. The method further comprises receiving a plurality of simulation action notifications from the computing system of the training environment. The method further comprises for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs and storing information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs. The method further comprises determining a set of completed BEEs. The set of completed BEEs comprising each active BEE that has all corresponding elicited actions met. The method further comprises, for each completed BEE of the set of completed BEEs, storing information regarding the training performance. The information comprises the one or more observable behaviors of the completed BEE.

[0005] Another example provides a computing system comprising a logic system operably coupled to a training environment that simulates a physical environment, and a computer-readable storage system. The computer-readable storage system comprises a behavioral elicitation event (BEE) database comprising a plurality of BEEs, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors, and instructions executable by the logic system. The instructions are executable to receive a simulation state update from a computing system of the training environment, query the BEE database to determine, from the plurality of BEEs, a set of active BEEs based upon the simulation state update, and receive a plurality of simulation action notifications. The instructions are further executable to, for each simulation action notification, determine whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs, and store information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs. The instructions are further executable to determine a set of completed BEEs. The set of completed BEEs comprises each active BEE that has all corresponding elicited actions met. The instructions are further executable to, for each completed BEE of the set of completed BEEs, store information regarding the completed BEE.
[0006] Another example provides a method of evaluating a training performance of a training subject against a competency framework. The training performance being performed in a training environment that simulates a physical environment. The method comprising receiving a first simulation state update from a computing system of the training environment, and querying a behavioral elicitation event (BEE) database to determine, from a plurality of BEEs stored in the BEE database, a first set of active BEEs based upon the first simulation state update. Each BEE of the plurality of BEEs comprises one or more elicited actions and one or more observable behaviors. The method further comprises receiving a plurality of simulation action notifications from the computing system of the training environment. The method further comprises, for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the first set of active BEEs, and storing information regarding the elicited action and corresponding observable behaviors as a response to the at least one active BEE in the first set of active BEEs. The method further comprises receiving a second simulation update from the computing system of the training environment, querying the BEE database to determine a second set of active BEEs based upon the second simulation state update, and comparing the first set of active BEEs to the second set of active BEEs. The method further comprises, for each active BEE in the first set of active BEEs and not in the second set of active BEEs, store the active BEE as an incomplete BEE and also store information regarding the incomplete BEE.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** shows a block diagram of an example computing system configured to utilize a BEE database to evaluate a training performance.
[0008] FIG. **2** shows a block diagram of an example BEE database used by the computing system of FIG. **1**.
[0009] FIG. **3** illustrates a block diagram of an example engine anti-ice BEE.
[0010] FIGS. **4**A and **4**B schematically illustrate an example algorithm that uses the BEE database of FIG. **1** to evaluate a training performance of a test subject.
[0011] FIG. **5** illustrates a table of example simulation updates corresponding to a successful completion of the engine anti-ice BEE of FIG. **3**
[0012] FIGS. **6** and **7** schematically illustrate example subprocesses of the algorithms of FIGS. **4**A

and **4**B.

[0013] FIG. **8** illustrates a flowchart illustrating an example method for evaluating a training performance of a training subject against a competency framework.

[0014] FIGS. **9-13** schematically show views of an example user interface (UI).

[0015] FIG. **14** shows a block diagram of an example computing system.

DETAILED DESCRIPTION

[0016] As previously mentioned, training of a pilot uses the competency-based training and assessment (CBTA) to evaluate a training subject (the pilot) against the ICAO competency framework. Current methods of using the CBTA in training solutions have an instructor evaluate the training subject on the ICAO competency framework by observing the training subject during a training performance, such as during training in a flight simulator. Further, the instructor must also provide supporting evidence showing how the training subject either successfully or unsuccessfully demonstrated observable behaviors corresponding with each competency in the ICAO competency framework. Such current methods of evaluation pose several problems. First, the ICAO competency framework comprises mostly soft skills. Soft skills are difficult for the instructor to objectively assess and provide feedback to the training subject upon successful completion of the ICAO competency framework. For example, evaluation by observation can be difficult to provide information with traceability between actions performed by the training subject and the ICAO competency framework. The lack of traceability can make it difficult to provide recommendations to help the training subject improve their training performance in a next training performance over a current training performance. Another problem with evaluation by observation is that a similar training performance can receive differing evaluations across different instructors. Additionally, the instructor needs to be present during the training performance for the observation. This can add cost to the pilot training to cover instructor pay and training.

[0017] Accordingly, examples are disclosed that relate to utilizing computer-implemented behavioral elicitation events (BEEs) to automate the evaluation of a training performance of a training subject against a competency framework. A BEE is an event that elicits particular behaviors out of the training subject to demonstrate specified soft skill competencies in the competency framework. The training performance is performed by the training subject in a training environment that simulates a physical environment. Example training environments are a fixed training device (FTD) flight simulator, a full motion simulator (FMS) flight simulator, and a virtual training environment. As described in more detail below, the disclosed examples utilize a BEE database comprising a plurality of BEEs to evaluate the training performance. Each BEE of the plurality of BEEs comprises one or more triggers, one or more elicited actions, and one or more observable behaviors, as discussed in more detail with reference to FIG. **2**. The disclosed examples receive simulation state updates from the training environment, and determine a set of active BEEs based on the simulation state update by querying the BEE database. The simulation state update indicates a change in a simulation state of the training environment.

[0018] Continuing, the disclosed examples receive a plurality of simulation action notifications from the training environment. The simulation action notifications indicate a corresponding plurality of user inputs received by the training environment. When an action represented by the simulation action notification matches an elicited action for an active BEE in the set of active BEEs, the disclosed examples store information regarding the elicited action and corresponding observable behaviors as a response to the active BEE. Further, when an active BEE has all corresponding elicited actions met, the disclosed examples determine that the active BEE is a completed BEE.

[0019] The disclosed examples further store information regarding any completed BEEs and/or incomplete BEEs. The information can be used as evidence supporting how the training subject demonstrated various competencies in the competency framework during the training performance. The information can also be used for instructional purposes as well as tailoring training scenarios for an individual training subject. Using the BEE database can provide scenario independent behavior detection for evaluation, which can help reduce author time of a training scenario over current methods where each scenario is authored independently of each other.

[0020] FIG. **1** depicts a block diagram of an example computing system **100** in communication with a training environment, which in this example takes the form of an aircraft training environment **102**. Here, aircraft training environment **102** is configured to simulate an aircraft environment for use in training pilots. In some examples, aircraft training environment **102** can be configured as a FTD, a FMS, or a virtual training environment. A training subject **104** performs a training performance in aircraft training environment **102** using a training interface **106** to interact with a training computing system **108**. Training computing system **108** is configured to perform a computer simulated training scenario. In some examples, training interface **106** and aircraft training environment **102** can be configured to simulate a flight deck within aircraft training environment **102** and to receive user inputs from training subject **104**. Examples user inputs include audio input, button presses or hardware selections into training interface **106**, video input, and other suitable inputs from training subject **104**.

[0021] In some examples, training computing system **108** can further be configured to utilize machine learning algorithms to analyze the various user inputs, such as using speech to text recognition to determine content of audio input or a classifier to identify specified gestures in the video input, for example. Further, the machine learning algorithms can be used with physiological measurements to help determine whether training subject **104** is paying attention to a desired portion of the flight deck in aircraft training environment **102**. Such a configuration can help to generate a perception model that can be used to detect elicited actions corresponding to various monitoring competencies in the competency framework, such as a cognitive workload management competency, for example. Training subject **104** further can enter user inputs in response to the computer simulated training scenario, such as a change in one or more of a simulated flight phase, simulated weather, or a state of a simulated aircraft, for example. In other examples, aircraft training environment **102** can have another configuration. In yet further examples, another training environment other than an aircraft training environment can be used.

[0022] Computing system **100** is configured to evaluate the training performance of training subject **104** against a competency framework, such as the ICAO pilot competencies, for example. Computing system **100** comprises a processor **110** operably coupled to aircraft training environment **102**, and a storage system **112**. Storage system **112** comprises a BEE database **114** comprising a plurality of BEEs as discussed with reference to FIG. **2**. Storage system **112** further comprises a BEE monitor **116** configured to be executable by processor **110**. Storage system **112** further comprises a learning record store (LRS) **118** configured to store information relating to the evaluation of the performance of training subject **104**. In some examples, LRS **118** can be configured to also store information from aircraft training environment **102**. In other examples, the information from a training environment and/or information relating to the performance evaluation can be stored in another location or across multiple locations. Further aspects of processor **110** and storage system **112** are discussed with reference to FIG. **9**. In other examples, computing system **100** can be configured to receive flight operations quality assurance (FOQA) data recorded during a live flight and to detect pilot behavior against the ICAO competency framework. Such a configuration can help to determine training recommendations for an airline fleet and/or to determine effectiveness of training programs of the airline fleet. While discussed herein with reference to aircraft training environment **102**, a BEE database can be used to evaluate another training performance against any suitable competency framework in other examples.

[0023] Each BEE stored in a BEE database indicates an event that elicits particular behaviors out of a training subject. FIG. **2** shows a block diagram of BEE database **114** in more detail. BEE database **114** comprises a first BEE **200**. First BEE **200** comprises one or more triggers **202** defining conditions when first BEE **200** is an active BEE. More specifically, each trigger can be an active trigger when a corresponding condition meets a simulation state of a training environment. In examples of pilot training, a trigger can comprise a condition of specified simulated air temperatures, specified flight phases of a simulated aircraft, and specified fuel levels, altitude, speed, and/or another status of the simulated aircraft. Further, examples of flight phases include standing, cruising, takeoff, landing, taxing, ascending, descending, and approach. Triggers **202** can comprise an optional timing condition **204** indicating a specified time to respond to triggers **202**. Timing condition **204** can also indicate whether training subject **104** is expected to respond before or wait to respond until after the specified time. As a specific example, a fuel-leak BEE has a trigger with a timing condition of wait until a desired time before taking an action to an increase in fuel flow. First BEE **200** further comprises one or more elicited actions **206** defining expected user inputs as a response to when first BEE **200** is an active BEE. Similar to triggers **202**, an elicited action **206** can comprise a timing condition **208**. In some examples, triggers **202** and/or elicited action **206** can comprise logic statements (not shown for clarity), such as Boolean logic (e.g., AND, OR, etc.), sequence statements, and conditional statements, for example. First BEE **200** further comprises one or more observable behaviors **210** indicating competencies demonstrated by training subject **104** by performing one or more elicited actions **206**. Further, each observable behavior **210** can line-up to a specified competency in a competency framework, such as specified pilot competencies in the ICAO competencies. A specific example of a BEE is provided in FIG. **3**.

[0024] Similarly, BEE database **114** comprises a second BEE **212** having one or more triggers, one or more elicited actions, and one or more observable behaviors. In such a manner, a BEE database can help to align elicited actions with observable behaviors for any suitable competency framework such that the BEE database can help to detect evidence of the observable behaviors in another simulated training environment. While FIG. **2** depicts two BEEs for clarity, BEE database **114** can comprise any suitable number of BEEs in other examples, such as a single BEE or more than two BEEs. BEE database **114** is illustrative. In other examples, BEE database **114** can comprise another configuration.

[0025] FIG. **3** depicts a block diagram of an example engine anti-ice BEE **300**. Engine anti-ice BEE **300** can be used to evaluate a training performance of a pilot in a training environment as discussed with reference in FIGS. **4**A and **4**B. In the example of FIG. **3**, engine anti-ice BEE **300** corresponds to an engine anti-ice scenario during a simulated standing flight phase in aircraft training environment **102**. As such, engine anti-ice BEE **300** comprises a flight phase indicator **302** of standing flight phase which indicates that engine anti-ice BEE **300** can be active during the simulated standing flight phase. Engine anti-ice BEE **300** further comprises a first trigger **304** configured to be an active trigger when a simulated air temperature of the training environment is under 10° C. Similarly, a second trigger **306** is configured to be an active trigger when the training environment has simulated visible air moisture.

[0026] Engine anti-ice BEE **300** further comprises a first elicited action (EA) **308** and a second elicited action **310**. Here, first elicited action **308** defines a user input to aircraft training environment **102** indicating that the pilot selected an engine anti-ice procedure in an electronic flag bag of aircraft training environment **102**. Second elicited action **310** defines a user input indicating that the pilot verbally communicated to their co-pilot that the engine anti-ice procedure was selected.

[0027] Engine anti-ice BEE **300** further comprises a first set of observable behaviors (OBs) **312** that define competencies related to demonstrated behaviors of the pilot performing first elicited action **308**. Here, first set of observable behaviors **312** indicates that a training subject demonstrated knowledge of procedures, knowing where to source required information, situational awareness, communication, problem solving, following standard operating procedures, and planning, prioritizing, and scheduling appropriate tasks. Similarly, engine anti-ice BEE **300** also comprises a second set of observable behaviors **314** related to the response to second elicited action **310**. Second set of observable behaviors **314** indicates that the training subject demonstrated conveying messages clearly, accurately, and concisely. First and second sets of observable behaviors **312**, **314** correspond to pilot competencies defined in the ICAO competencies. FIG. **3** is illustrative. In other examples, engine anti-ice BEE **300** can have another configuration.

[0028] One or more computing systems can be configured to perform an algorithm that uses a BEE database to evaluate a training performance of a training subject in a training environment, such as aircraft training environment **102**, for example. FIGS. **4**A and **4**B illustrate overview flowcharts of an example algorithm **400** that uses BEE database **114**. Here, a computing system **402** and computing system **100** perform algorithm **400**. In some examples, computing system **402** can be a part of training computing system **108**. Computing system **402** comprises a learning record store (LRS) **404** configured to store information regarding user inputs from the training subject, such as interactions with another individual participating in the training environment, and information regarding actions performed by the training subject, for example. LRS **404** is further configured to store information about a simulated state of the training environment, such as simulated weather, for example. Computing system **402** further comprises an experience application programming interface (xAPI) monitor **406** and an xAPI logger **408**. Briefly, xAPI monitor **406** is configured to monitor LRS **404** for any new statements received from the training environment. xAPI logger **408** is configured to transmit information from BEE monitor **116** to LRS **404** to be stored. In other examples, another configuration of one or more computing systems comprising a BEE database can perform algorithm **400**. For example, other data sources than the LRS can be used for storing, writing and/or receiving data for use in a simulation evaluation.

[0029] With reference to FIG. **4**A, xAPI monitor **406** queries, at **410**, a session in LRS **404** for an update statement comprising a simulation state update or simulation action notification. In response, LRS **404** returns, at **412**, the update statement to xAPI monitor **406**. At **414**, xAPI monitor **406** sends the update statement to BEE monitor **116**. Alternatively, in the example of FIG. **4**B, the querying, at **410**, for update statements and the returning, at **412**, of the update statements are omitted.

[0030] After BEE monitor **116** receives the update statement and when the update statement comprises a simulation action notification, as indicated at **416**, algorithm **400** moves to corresponding subprocesses as discussed with reference to FIG. **7**. Alternatively, when the statement update comprises a simulation state update, as indicated at **418**, algorithm **400** moves to corresponding subprocesses discussed with reference to FIG. **6**. The examples of FIG. **4**A or FIG. **4**B can be selected based on a configuration of the training environment. For example, the example of FIG. **4**A can be selected for use with a FTD or FMS. The example of FIG. **4**B can be selected for use with a virtual training environment. FIGS. **4**A and **4**B are illustrative. In other examples, algorithm **400** can use another overview flowchart.

[0031] For illustration, subprocesses of algorithm **400** are discussed with reference to engine anti-ice BEE **300**. As such, FIG. **5** depicts a table of example update statements **500** corresponding to a successful performance of training subject **104** in response to engine anti-ice BEE **300**. Here, a first simulation state update **502** comprises a change in simulated weather reflecting that the simulated air temperature is 0° C. Further, a second simulation state update **504** comprises a change in the simulated weather reflecting that there is simulated visible air moisture. Next, a first simulation action notification **506** indicates a corresponding user input of selecting an engine anti-ice procedure in an electronic flight bag of aircraft training environment **102**. Further, a second simulation action notification **508** indicates a corresponding user input of communicating that the engine anti-ice procedure was selected. Such a user input can comprise audio input, for example. In other examples, algorithm **400** can use another BEE and/or other update statements.

[0032] As previously mentioned, algorithm **400** is discussed using the example of engine anti-ice BEE **300** during a simulated standing flight phase. In the current example, algorithm **400** will perform more than one loop through various subprocesses. During each loop, algorithm **400** performs different subprocesses and/or skips different subprocesses. A first loop begins when xAPI monitor **406** sends, at **414**, first simulation state update **502** to BEE monitor **116**, and algorithm **400** moves to FIG. **6** as discussed above. With reference to FIG. **6**, BEE monitor **116** queries, at **600**, BEE database **114** for any triggers related to first simulation state update **502** (sim. air temp. is 0° C.). In response, BEE database **114**, returns, at **602**, first trigger **304** (air temp. under 10° C.) of engine anti-ice BEE **300**. Here, first trigger **304** is an active trigger. Therefore, BEE monitor **116** stores, at **604**, first trigger **304** in a set of active triggers. In this example, no active triggers have become inactive triggers based on first simulation state update **502**, so BEE monitor **116** skips **606**. In other examples, BEE monitor **116** can remove, at **606**, any inactive triggers from the set of active triggers. An inactive trigger is a trigger having a condition that is not met by a current state of the training environment. As a specific example, a change in a simulated flight phase can cause one or more triggers to become inactive. In such a manner, the set of active triggers can help to enable a quick query for any active BEEs.

[0033] Continuing, BEE monitor **116** queries, at **608**, BEE database **114** for any active BEEs based on the set of active triggers. As previously mentioned, an active BEE has all corresponding triggers as active triggers. Here, engine anti-ice BEE **300** has one active trigger (first trigger **304**) and one inactive trigger (second trigger **306**). As a result, BEE database **114** sends, at **610**, in response to the query, no active BEEs to BEE monitor **116**. Further, as no BEEs are in a prior set of active BEEs and not in the set of active triggers resulting from first trigger **304** being in the set of active triggers, BEE monitor **116** skips **612** and returns to **420**.

[0034] In other examples when a BEE is in the prior set of active BEEs and not in the set of active BEEs, BEE monitor **116** determines, at **612**, a set of incomplete BEEs comprising one or more BEEs that are in the prior set of active BEEs and not in the set of active BEEs. For each incomplete BEE in the set of incomplete BEEs, algorithm **400** can store information regarding the incomplete BEE. Specifically, BEE monitor **116** queries BEE database **114** for observable behaviors of the incomplete BEE for each corresponding trigger having an optional timing condition at **614**, and/or for each corresponding elicited action at **616**. In response, BEE database **114** returns the corresponding observable behaviors of the incomplete BEE at **618** and at **620**, respectively. Next, BEE monitor **116** sends, at **622**, an observable behavior evaluation for each incomplete BEE based on the corresponding observable behaviors of the incomplete BEE to xAPI logger **408**. The observable behavior evaluation can indicate that a subset of the corresponding observable behaviors was demonstrated and/or partially demonstrated in response to the incomplete BEE. As a specific example, the evaluation can indicate that two elicited actions were completed, but there was an opportunity to complete five elicited actions. Continuing at **624**, xAPI logger **408** then sends information regarding the observable behaviors demonstrated, partially demonstrated, and/or failed to be demonstrated to be stored to LRS **404**. In such a manner, LRS **404** can store information regarding completed elicited actions, and also information regarding missed elicited actions in response to an incomplete BEE.

[0035] Returning to the example of engine anti-ice BEE **300**, algorithm **400** continues to a second loop when xAPI monitor **406** sends, at **414**, second simulation state update **504** (visible moisture in the air). BEE monitor **116** skips **416** and moves to the subprocess related to a simulation state update at **418**. In FIG. **6**, BEE monitor **116** again queries, at **600**, BEE database **114** for any triggers related to second simulation state update **504**. Here, BEE monitor **116**, receives, at **602**, second trigger **306** (simulated visible air moisture) and stores, at **604**, second trigger **306** in the set of active triggers. BEE monitor **116** further queries, at **608**, BEE database **114** for any active BEEs based on the set of active triggers. Here, first trigger **304** (weather is under 10° C.) and second trigger **306** (visible air moisture) of engine anti-ice BEE **300** are active. Thus, BEE database **114** stores engine anti-ice BEE **300** in a set of active BEEs. Similar to the first loop, no BEEs have become incomplete BEEs as a result of second simulation state update **504** and as such, algorithm **400** returns to **420**.

[0036] In the next, a third loop of algorithm **400**, xAPI monitor **406** sends, at **414**, first simulation action notification **506** (the user input of selecting the engine anti-ice procedure). Here, an action represented by first simulation action notification **506** matches first elicited action **308** of engine anti-ice BEE **300**, and as such, algorithm **400** moves to 7 at **416**. Now with reference to FIG. **7**, first elicited action **308** does not have a corresponding timing condition. Similarly, no triggers of engine anti-ice BEE **300** has an optional timing condition. Thus, algorithm **400** skips **700** and **702**.

[0037] In other examples, an elicited action, as indicated at **704**, and/or a corresponding trigger, as indicated at **706**, can have an optional timing condition. In such examples, BEE monitor **116** checks, at **700**, whether a timing of the simulation action notification is within the timing condition of the elicited action. As a specific example, a fuel-leak BEE has a fuel-flow trigger of an increased fuel flow indication with a timing condition. Further, the timing condition indicates that a specified time duration elapses before performing a corresponding elicited action in response to the fuel-flow trigger. BEE monitor **116** further checks, at **702**, whether a timing of a trigger corresponding to the elicited action. Information regarding the timing of the simulation action notification and/or the timing of the trigger can be included in a BEE evaluation message to be stored.

[0038] Returning to the example of engine anti-ice BEE **300**, BEE monitor **116** queries, at **708**, BEE database **114** for one or more observable behaviors (OBs) corresponding to first elicited action **308**. BEE monitor **116** receives, at **710**, first set of observable behaviors **312**. Further, BEE monitor **116** stores information regarding first elicited action **308** as a response to engine anti-ice BEE **300** by sending, at **712**, a BEE evaluation to xAPI logger **408**, and in response, xAPI logger **408** sends, at **714**, a responded message to be stored to LRS **404**. The information can comprise first set of observable behaviors **312** as behaviors demonstrated by the training subject performing first elicited action **308**, for example. In other examples, other information regarding an elicited action as a response to an active BEE can be stored in another suitable manner.

[0039] Continuing the current example, engine anti-ice BEE **300** has first elicited action **308** met and second elicited action **310** not met. Therefore, at least one corresponding elicited action of engine anti-ice BEE **300** is not met, and therefore engine anti-ice BEE **300** is still an active BEE. Thus, algorithm **400** skips **716** and moves to querying, at **600**, for active triggers in FIG. **6**. In this example, first simulation action notification **506** (user input of selecting engine anti-ice procedure) does not match a condition for another trigger, and thus BEE database **114** does not return any new active triggers. Further, no new BEEs become active BEEs, and as such algorithm **400** returns to **420**.

[0040] Similarly, algorithm **400** starts a fourth loop when BEE monitor **116** receives, at **414**, second simulation action notification **508** and moves to **416**. In FIG. **7**, BEE monitor **116** queries, at **708**, for observable behaviors (OBs) corresponding to second elicited action **310**, and receives, at **710**, second set of observable behaviors **314**. Further, BEE monitor **116** stores information regarding second elicited action **310** as a response to engine anti-ice BEE **300** using xAPI logger **408** as described above with reference to first elicited action **308**.

[0041] Here in the current example, engine anti-ice BEE **300** has all corresponding elicited actions met (e.g., both first and second elicited actions **308**, **310**). Thus, BEE monitor **116** determines, at **716**, that engine anti-ice BEE **300** is a completed BEE. As previously discussed, engine anti-ice BEE **300** has no triggers having a timing condition, and therefore BEE monitor **116** skips **718**. Next, BEE monitor **116** queries, at **720**, for observable behaviors that are demonstrated by simply performing each corresponding elicited action of a completed BEE. In the current example of engine anti-ice BEE **300**, BEE database **114** returns, at **722**, first set of observable behaviors **312** for first elicited action **308**, and second set of observable behaviors **314** for second elicited action **310**. In response, BEE monitor **116** sends, at **724**, an observable behavior evaluation message to xAPI logger **408** for each observable behavior in first and second sets of observable behaviors **312**, **314**. At **726**, xAPI logger **408** sends an observable behavior demonstrated message to be stored to LRS **404**. In such a manner, the observable behaviors demonstrated by the pilot in context of engine anti-ice BEE **300** are stored. In other examples, a BEE can have a trigger with a timing condition, as indicated at **718**. In such examples, BEE monitor **116** queries, at **728**, for observable behaviors that are demonstrated because of a timing of a corresponding elicited action relative to a timing of the trigger of the BEE. At **730**, BEE monitor **116** receives the observable actions and further includes information regarding the timings in the observable behavior evaluation sent at **724**.

[0042] Similar to the third loop, algorithm **400** next moves to query triggers at **600** in FIG. **6**. As no new triggers have become active resulting from the training subject performing second elicited action **310** or the completion of engine anti-ice BEE **300**, algorithm **400** skips through and returns to **420** and waits for a new simulation update. In such a manner, algorithm **400** utilizing BEE database **114** can evaluate the training performance of the pilot against the ICAO competencies. FIGS. **6** and **7** are illustrative. While algorithm **400** and corresponding subprocesses are discussed with reference to evaluating the pilot against engine anti-ice BEE **300**, algorithm **400** can be used to evaluate a training performance of another training subject against another competency framework in other examples. Similarly in further examples, another algorithm utilizing BEE database **114** can be used to evaluate any training subject against any suitable competency framework performed in any suitable training environment that simulates a physical environment.

[0043] FIG. **8** illustrates a flowchart of an example method **800** for evaluating a training performance of a training subject against a competency framework. The training performance is performed in a training environment that simulates a physical environment, such as aircraft training environment **102**, for example. In such an example the competency framework can comprise the ICAO competency framework, for example. In other examples, another training environment and/or another competency framework can be used. Method **800** can be performed by any suitable computing system comprising a BEE database, such as computing system **100**, for example.

[0044] Method **800** comprises, at **802**, receiving a simulation state update from a computing system of the training environment, such as training computing system **108**, for example. In examples simulating an aircraft environment, the simulation state update can comprise a change in one or more of a simulated flight phase, simulated weather, or a state of a simulated aircraft. In other examples, the simulation state update can comprise another change in a simulation state the physical environment (e.g., simulated weather, simulated location, simulated alarms, etc.).

[0045] Returning, method **800** comprises, at **804**, querying a BEE database to determine, from a plurality of BEEs stored in the BEE database, a set of active BEEs based upon the simulation state update. BEE database **114** can be used, for example. As discussed above, each BEE of the plurality of BEEs comprises one or more triggers, one or more elicited actions, and one or more observable behaviors. Querying the BEE database to determine the set of active BEEs comprises querying the BEE database for one or more triggers that have a condition matching the simulation state update, at **806**. Determining the set of active BEEs also comprises storing the one or more triggers as a set of active triggers, at **808**. Further, determining the set of active BEEs comprises, at **810**, determining one or more inactive triggers in the set of active triggers by comparing a corresponding condition for each active trigger in the set of active triggers to the simulation state update, and removing the one or more inactive triggers from the set of active triggers. Determining the set of active BEEs further comprises, at **811**, querying the BEE database to determine whether any BEEs are active BEEs based on the set of active triggers. In other examples, the set of active BEEs can be determined in another suitable manner.

[0046] Method **800** further comprises, at **812**, receiving a plurality of simulation action notifications from the computing system of the training environment. The plurality of simulation action notifications indicates a corresponding plurality of user inputs received by the training environment. Examples of user inputs include detected speech, a selected input on a user interface of the training environment, detected eye movement, and/or detected gestures. In other examples, method **800** can again query the BEE database to determine a set of active BEEs as discussed with reference to **804**. Method **800** further comprises, at **814**, for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs. Method **800** comprises, when the action represented by the simulation action notification matches the elicited action, determining whether the elicited action has a timing condition, and storing information regarding a timing of the simulation action notification at **816**. The timing condition can indicate that a desired timing of the simulation action notification is after or at a desired time duration. Alternatively, the timing condition can indicate that a desired timing of the simulation action notification is within a desired time duration. Method **800** further comprises, at **818**, comparing the timing of the simulation action notification with a timing of a corresponding trigger of the at least one active BEE, and storing information regarding the comparison of the timings. In other examples, method **800** can omit **816** and/or **818**.

[0047] Method **800** comprises, at **820**, storing information regarding the elicited action and corresponding observable behaviors as a response to the at least one active BEE in the set of active BEEs. Method **800** further comprises, at **822**, comparing a prior set of active BEEs to the set of active BEEs and for each active BEE in the prior set of active BEEs and not in the set of active BEEs, storing the active BEE as an incomplete BEE. Method **800** comprises, at **824**, storing information regarding the training performance. The information comprises one or more missed elicited actions in response to the incomplete BEE. In other examples, **822** and **824** may be omitted.

[0048] Method **800** comprises, at **826**, determining a set of completed BEEs. The set of completed BEEs comprises each active BEE that has all corresponding elicited actions met. At **828**, method **800** further comprises, for each completed BEE of the set of completed BEEs, storing information regarding the training performance. The information comprises the one or more observable behaviors of the completed BEE. In some examples, the information can further comprise any suitable information regarding the completed BEE, such as timing information, for example.

[0049] Using a BEE database for an evaluation of a training performance as described herein can provide various benefits. The evaluation can provide data on the training performance to help an instructor objectively assess soft skill competencies and increase consistency of inter-rater reliability over evaluations by observation of the instructor. The evaluation can also help to provide detailed evidence for use when asserting the proficiency of a training subject performing the training performance. Further, the evaluation can help to provide traceability between elicited actions and specified competencies in a competency framework. Such traceability can help to provide feedback to the training subject regarding recommendations to raise a competency level during a next training performance over a current training performance. Additionally, the evaluation can help to enable tailoring a follow-on training scenario for the training subject based on the current training performance and/or prior training performances.

[0050] Additionally or alternatively, using the BEE database for evaluations across a plurality of training performances can provide data to enable analytics across training subjects to determine recommendations for updating a training scenario used in a training environment. Further, the data can also help to evaluate effectiveness of training environments, specifically virtual training environments. Similarly, such evaluations can help to analyze operational data to determine effectiveness of training scenarios and/or to determine recommended training scenarios across training subjects.

[0051] A user interface (UI) can be used to display information stored in the BEE database to a user, for example, to facilitate evaluation of a training performance of a training subject during a simulated training scenario. Example users include a training subject and/or an instructor. FIGS. **9-13** schematically show various interfaces displayed by an example UI **900**. In this example, the UI **900** displays information acquired during a flight simulation for a civilian aircraft. However, a UI according to the present disclosure can display information acquired during other simulated training scenarios in other examples.

[0052] Referring first to FIG. **9**, UI **900** comprises a menu having menu items in the form of a series of tabs **901** associated with respective flight phases of the simulated flight, such as the pre-flight, taxi, takeoff, cruise, descent, landing, and taxi phases. The user can select any tab (or other form of menu item) of the series of tabs **901** to view information relating to the selected flight phase of the simulated flight on the UI **900** for the selected tab. Other simulations than a civilian flight simulation also can have multiple phases that can be displayed using tabs on a user interface. Further, in other examples, a flight phase menu (or menu of phases of any other suitable simulation) can be displayed in any other suitable format, such as in a drop-down menu.
[0053] FIG. **9** shows an example of UI **900** wherein the “takeoff” tab of the series of tabs **901** is selected. When the “takeoff” tab is selected, the UI **900** displays a graph **902** depicting a simulated flight path associated with the takeoff phase of the simulated flight. The graph **902** displays an altitude and/or other positional information of the simulated aircraft as a function of time. The “takeoff” display of the UI **900** also comprises an “environmentals” display **904**. Environmentals display **904** displays a simulated state of the training environment, such as wind direction, wind speed, and temperature.
[0054] The UI **900** further comprises expandable displays **906**A and **906**B that display example competencies, observable behaviors, and/or other information for the takeoff phase of the simulated flight. More particularly, the expandable display **906**A comprises information regarding a communication observable behavior, and the expandable display **906**B comprises information regarding an application of procedures observable behavior. The expandable displays can be selected by a user to cause display of an expanded observable behavior display. The UI **900** further comprises a display of example elicited actions **908** relating to the observable behaviors displayed in expandable displays **906**A and **906**B. Example elicited actions relating to the communication observable behavior comprise “called 80 ‘knots’” and “call ‘rotate’”. Example elicited actions relating to the application of procedure observable behavior comprise “set landing gear lever to UP” and “verified VNAV engaged”. The competency of the training subject during the takeoff phase of the simulated flight can be evaluated based on the completion of the elicited actions.
[0055] The UI **900** further comprises completion indicators **910** corresponding to each elicited action. As shown in FIG. **9**, the completion indicator for “call ‘rotate’” is shown in dashed lines, and the corresponding elicited action display reads “did not call ‘rotate’”, indicating that this elicited action relating to the communication observable behavior was not completed by the training subject in the takeoff phase of the simulated flight. Expandable displays **906**A and **906**B can be selected to display further detail regarding the completion of elicited actions corresponding to the observable behaviors.
[0056] FIG. **10** shows an example screen of UI **900** that can be displayed when the expandable display **906**A for the communication observable behavior is selected. The UI **900** comprises a progress bar **912**A displayed within the selected expandable display **906**A. Progress bar **912**A reflects a level of completion of elicited actions relating to the communication observable behavior. Details of each elicited action relating to the communication observable behavior are displayed at **914**. Each expected elicited action is displayed at **916**, and the performance of a training subject relating to each elicited action is displayed at **918**. In this example, additional details such as time also are displayed. One example expected elicited action is “call ‘rotate’”, displayed at **920**. The corresponding performance for this expected elicited action is displayed at **922**, and reads “did not call ‘rotate’”, reflecting that the training subject did not perform the expected elicited action. Thus, the progress bar **912**A is displayed as incomplete. A user can navigate UI **900** in this manner to review all details relating to the communication observable behavior and the corresponding elicited actions taken during the takeoff phase simulated flight in expandable display **906**A. The expandable display **906**B comprising the application of procedures observable behavior is unselected, as indicated by the shading in FIG. **10**.
[0057] FIG. **11** shows an example screen of the UI **900** that can be displayed when the expandable

display **906**B for the application of procedures observable behavior is selected. The UI **900** comprises a progress bar **912**B displayed within the selected expandable display **906**B. Progress bar **912**B reflects a level of completion of elicited actions relating to the application of procedures observable behavior. Progress bar **912**B is complete, and thus reflects a successful completion of elicited actions relating to the application of procedures observable behavior. Details of each elicited action are displayed at **924**. The expected elicited action is displayed at **926**, and the performance of the training subject is displayed at **928**. Further details, such as time, also are displayed. One example expected elicited action comprises "verify VNAV engaged", displayed at **930**. The corresponding performance for this expected elicited action is displayed at **932**, and reads "correctly verified VNAV engaged", reflecting the performance of the training subject. In the example shown in FIG. **9**C, all elicited actions were completed by the training subject. Thus, progress bar **912**B is displayed as complete. In this manner, a user can select expandable display **906**B to review all details relating to the application of procedures observable behavior and the corresponding elicited actions taken during the takeoff phase of the simulated flight.

[0058] The user also may navigate to other tabs of UI **900** to review training data relating to other flight phases of the simulated flight. FIG. **12** shows an example of UI **900** wherein the "cruise" tab of the series of tabs **901** is selected. The "cruise" tab comprises an image **934** depicting a simulated flight path associated with a cruise phase of the simulated flight. Image **934** displays an overhead map view of the simulated aircraft. Image **934** also can display simulated obstacles, such as terrain features **935** and/or other aircraft **936**. Image **934** also can display simulated altitudes of aircraft displayed in image **934**, and/or any other suitable information for the cruise phase of the simulated flight.

[0059] In the example shown in FIG. **12**, the simulated flight comprises a flight event, displayed in image **934** as a line **937**. The flight event can be a trigger determining one or more active BEEs. For example, the flight event displayed at **937** can be a fuel leak which determines an active fuel-leak BEE. Thus, competencies of the training subject relating to a fuel leak can be measured. The "cruise" tab of the UI **900** also comprises the "environmentals" display **904**. As mentioned above, an update in the simulated state of the training environment can be a trigger determining one more active BEEs. A BEE can comprise one or more elicited actions defining expected user inputs as a response to when the BEE is an active BEE. The BEE further comprises one or more observable behaviors indicating competencies demonstrated by a training subject by performing the one or more elicited actions.

[0060] Similarly to the "takeoff" tab described in FIG. **9**, the UI **900** in FIG. **12** comprises expandable displays of example competencies, observable behaviors, and/or other information for the cruise phase of the simulated flight. The expandable display **938**A comprises a problem solving and decision making—seeks accurate and adequate information from appropriate sources observable behavior, expandable display **938**B comprises a situation awareness—develops effective contingency plans based upon potential risks associated with threats and errors observable behavior, and expandable display **938**C comprises an application of procedures—applies relevant operating instructions, procedures, and techniques in a timely manner observable behavior.

[0061] The UI **900** further comprises a display of elicited actions **940** relating to the observable behaviors in expandable displays **938**A, **938**B, and **938**C. For example, elicited actions relating to the problem solving and decision making observable behavior displayed at **938**A can comprise "checked weather for nearest suitable airport" and "checked performance calculations for nearest suitable airport". Example elicited actions relating to the situational awareness observable behavior displayed at **938**B can comprise "diverted to nearest suitable airport." Example elicited actions relating to the application of procedures observable behavior in response to the flight event displayed at **937** can comprise "opened fuel leak engine checklist" and "navigated to FUEL page." In other examples, any other suitable elicited actions can be displayed. UI **900** further comprises completion indicators **942** corresponding to each elicited action displayed in the display of elicited actions **940**. As shown in FIG. **12**, all example elicited actions relating to the example observable behaviors of expandable displays **938**A, **938**B, and **938**C are successfully completed. As mentioned above with regard to the "takeoff" tab of UI **900**, expandable displays **938**A, **938**B, and **938**C can be expanded to display detailed data regarding observable behaviors and the corresponding elicited actions taken during the "cruise" phase of the simulated flight.

[0062] The UI **900** further comprises a "recap" tab configured to display summarized information regarding training performance. FIG. **13** shows an example of the UI **900** wherein the "recap" tab is selected. The "recap" tab comprises a bar graph display **944** summarizing scores of a plurality of tests. The scores relate to successfully completed BEEs completed by the training subject during the simulated flight. In this manner, the UI **900** can display information enabling the user to determine competencies of the test subject. While the examples of FIGS. **9-12** show UI screen for the takeoff phase, cruise phase, and recap functionality, it will be understood that UI screens for other phases can contain any other suitable information for the relevant phases.

[0063] In some examples the methods and processes described herein can be tied to a computing system of one or more computing devices. In particular, such methods and processes can be implemented in hardware as described above, as a computer-application program or service, an application-programming interface (API), a library, and/or other computer-program product.

[0064] FIG. **14** schematically shows a simplified representation of a computing system **1400** configured to provide any to all of the compute functionality described herein. Computing system **1400** can take the form of one or more personal computers, server computers, and computers integrated with aircraft, as examples. Computing system **100**, training computing system **108**, and computing system **402** are examples of computing system **1400**.

[0065] Computing system **1400** includes a logic subsystem **1402** and a storage subsystem **1404**. Computing system **1400** can optionally include a display subsystem **1406**, input subsystem **1408**, communication subsystem **1410**, and/or other subsystems not shown in FIG. **14**.

[0066] Logic subsystem **1402** includes one or more physical devices configured to execute instructions. For example, the logic subsystem **1402** can be configured to execute instructions that are part of one or more applications, services, programs, routines, libraries, objects, components, data structures, or other logical constructs. Such instructions can be implemented to perform a task, implement a data type, transform the state of one or more components, achieve a technical effect, or otherwise arrive at a desired result.

[0067] The logic subsystem **1402** can include one or more hardware processors configured to execute software instructions. Additionally, or alternatively, the logic subsystem **1402** can include one or more hardware or firmware devices configured to execute hardware or firmware instructions. Processors of the logic subsystem **1402** can be single-core or multi-core, and the instructions executed thereon can be configured for sequential, parallel, and/or distributed processing. Individual components of the logic subsystem **1402** optionally can be distributed among two or more separate devices, which can be remotely located and/or configured for coordinated processing. Aspects of the logic subsystem **1402** can be virtualized and executed by remotely accessible, networked computing devices configured in a cloud-computing configuration.

[0068] Storage subsystem **1404** includes one or more physical devices configured to temporarily and/or permanently hold computer information such as data and instructions executable by the logic subsystem. When the storage subsystem **1404** includes two or more devices, the devices can be collocated and/or remotely located. Storage subsystem **1404** can include volatile, nonvolatile, dynamic, static, read/write, read-only, random-access, sequential-access, location-addressable, file-addressable, and/or content-addressable devices. Storage subsystem **1404** can include removable and/or built-in devices. When the logic subsystem executes instructions, the state of storage subsystem **1404** can be transformed—e.g., to hold different data.

[0069] Storage subsystem **1404** can include removable and/or built-in devices. Storage subsystem **1404** can include optical memory (e.g., CD, DVD, HD-DVD, Blu-Ray Disc, etc.), semiconductor memory (e.g., RAM, EPROM, EEPROM, etc.), and/or magnetic memory, among others. Storage subsystem **1404** can include volatile, nonvolatile, dynamic, static, read/write, read-only, random-access, sequential-access, location-addressable, file-addressable, and/or content-addressable devices.

[0070] Aspects of logic subsystem **1402** and storage subsystem **1404** can be integrated together into one or more hardware-logic components. Such hardware-logic components can include program-and application-specific integrated circuits (PASIC/ASICs), program- and application-specific standard products (PSSP/ASSPs), system-on-a-chip (SOC), and complex programmable logic devices (CPLDs), for example.

[0071] The logic subsystem **1402** and the storage subsystem can cooperate to instantiate one or more logic machines. As used herein, the term "machine" is used to collectively refer to the combination of hardware, firmware, software, instructions, and/or any other components cooperating to provide computer functionality. In other words, "machines" are never abstract ideas and always have a tangible form. A machine can be instantiated by a single computing device, or a machine can include two or more sub-components instantiated by two or more different computing devices. In some implementations a machine includes a local component (e.g., software application executed by a computer processor) cooperating with a remote component (e.g., cloud computing service provided by a network of server computers). The software and/or other instructions that give a particular machine its functionality can optionally be saved as one or more unexecuted modules on one or more suitable storage devices.

[0072] When included, display subsystem **1406** can be used to present a visual representation of data held by storage subsystem **1404**. This visual representation can take the form of a graphical user interface (GUI). As the herein described methods and processes change the data held by the storage subsystem, and thus transform the state of the storage subsystem, the state of display subsystem **1406** can likewise be transformed to visually represent changes in the underlying data. Display subsystem **1406** can include one or more display devices utilizing virtually any type of technology. Such display devices can be combined with the logic subsystem and the storage subsystem in a shared enclosure, or such display devices can be peripheral display devices.

[0073] When included, input subsystem **1408** can comprise or interface with one or more input devices such as a keyboard and touch screen. In some examples, the input subsystem can comprise or interface with selected natural user input (NUI) componentry. Such componentry can be integrated or peripheral, and the transduction and/or processing of input actions can be handled on- or off-board. Example NUI componentry can include a microphone for speech and/or voice recognition; and an infrared, color, stereoscopic, and/or depth camera for machine vision and/or gesture recognition.

[0074] When included, communication subsystem **1410** can be configured to communicatively couple computing system **1400** with one or more other computing devices. Communication subsystem **1410** can include wired and/or wireless communication devices compatible with one or more different communication protocols. As non-limiting examples, the communication subsystem can be configured for communication via a wireless telephone network, or a wired or wireless local- or wide-area network. In some examples, the communication subsystem may allow computing system **1400** to send and/or receive messages to and/or from other devices via a network such as the Internet.

[0075] Further, the disclosure comprises configurations according to the following clauses.

[0076] Clause 1. A method of evaluating a training performance of a training subject against a competency framework, the training performance being performed in a training environment that simulates a physical environment, the method comprising: receiving a simulation state update from a computing system of the training environment; querying a behavioral elicitation event (BEE) database to determine, from a plurality of BEEs stored in the BEE database, a set of active BEEs based upon the simulation state update, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors; receiving a plurality of simulation action notifications from the computing system of the training environment; for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs and storing information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs; determining a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met; and for each completed BEE of the set of completed BEEs, storing information regarding the training performance, the information comprising the one or more observable behaviors of the completed BEE.

[0077] Clause 2. The method of clause 1, wherein each BEE of the plurality of BEEs further comprises one or more triggers, and wherein querying the BEE database to determine the set of active BEEs comprises, querying the BEE database for one or more triggers that have a condition matching the simulation state update, storing the one or more triggers as a set of active triggers, and querying the BEE database to determine whether any BEEs are active BEEs based on the set of active triggers.

[0078] Clause 3. The method of clause 2, wherein querying the BEE database to determine the set of active BEEs further comprises determining one or more inactive triggers in the set of active triggers by comparing a corresponding condition for each active trigger in the set of active triggers to the simulation state update, and removing the one or more inactive triggers from the set of active triggers.

[0079] Clause 4. The method of clause 1, further comprising, comparing a prior set of active BEEs to the set of active BEEs, and for each active BEE in the prior set of active BEEs and not in the set of active BEEs, storing the active BEE as an incomplete BEE and also storing information regarding the training performance, the information comprising one or more missed elicited actions in response to the incomplete BEE.

[0080] Clause 5. The method of clause 1, further comprising, when the action represented by the simulation action notification matches the elicited action for the at least one active BEE in the set of active BEEs, determining whether the elicited action has a timing condition, and storing information regarding a timing of the simulation action notification.

[0081] Clause 6. The method of clause 5, further comprising comparing the timing of the simulation action notification with a timing of a corresponding trigger of the at least one active BEE, and storing information regarding the comparison of the timings.

[0082] Clause 7. The method of clause 1, wherein the simulation state update comprises a change in one or more of a simulated flight phase, simulated weather, or a state of a simulated aircraft.

[0083] Clause 8. The method of clause 1, wherein the plurality of simulation action notifications indicates a corresponding plurality of user inputs received by the training environment.

[0084] Clause 9. A computing system comprising: a logic system operably coupled to a training environment that simulates a physical environment; and a computer-readable storage system comprising, a behavioral elicitation event (BEE) database comprising a plurality of BEEs, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors, and instructions executable by the logic system to receive a simulation state update from a computing system of the training environment, query the BEE database to determine, from the plurality of BEEs, a set of active BEEs based upon the simulation state update, receive a plurality of simulation action notifications, for each simulation action notification, determine whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs, and store information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs, determine a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met, and for each completed BEE of the set of completed BEEs, store information regarding the completed BEE.

[0085] Clause 10. The computing system of clause 9, wherein each BEE of the plurality of BEEs further comprises one or more triggers, and wherein the instructions executable to query the BEE database to determine the set of active BEEs comprise instructions executable to query the BEE database for one or more triggers that have a condition matching the simulation state update, store the one or more triggers as a set of active triggers, and query the BEE database to determine whether any BEEs are active BEEs based on the set of active triggers.
[0086] Clause 11. The computing system of clause 10, wherein the instructions executable to query the BEE database to determine the set of active BEEs further comprise instructions executable to determine one or more inactive triggers in the set of active triggers by comparing a corresponding condition for each active trigger in the set of active triggers to the simulation state update, and to remove the one or more inactive triggers from the set of active triggers.
[0087] Clause 12. The computing system of clause 9, wherein the instructions are further executable to compare a prior set of active BEEs to the set of active BEEs, and for each active BEE in the prior set of active BEEs and not in the set of active BEEs, store the active BEE as an incomplete BEE and also store information regarding the incomplete BEE.
[0088] Clause 13. The computing system of clause 9, wherein the instructions are further executable to, when the action represented by the simulation action notification matches the elicited action for the at least one active BEE in the set of active BEEs, determine whether the elicited action has a timing condition, and store information regarding a timing of the simulation action notification.
[0089] Clause 14. The computing system of clause 9, wherein the simulation state update comprises a change in one or more of a simulated flight phase, simulated weather, or a state of a simulated aircraft.
[0090] Clause 15. The computing system of clause 9, wherein the plurality of simulation action notifications indicates a corresponding plurality of user inputs received by the training environment.
[0091] Clause 16. A method of evaluating a training performance of a training subject against a competency framework, the training performance being performed in a training environment that simulates a physical environment, the method comprising: receiving a first simulation state update from a computing system of the training environment; querying a behavioral elicitation event (BEE) database to determine, from a plurality of BEEs stored in the BEE database, a first set of active BEEs based upon the first simulation state update, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors; receiving a plurality of simulation action notifications from the computing system of the training environment; for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the first set of active BEEs and storing, information regarding the elicited action and corresponding observable behaviors as a response to the at least one active BEE in the first set of active BEEs; receiving a second simulation state update from the computing system of the training environment; querying the BEE database to determine a second set of active BEEs based upon the second simulation state update; comparing the first set of active BEEs to the second set of active BEEs; and for each active BEE in the first set of active BEEs and not in the second set of active BEEs, store the active BEE as an incomplete BEE and also store information regarding the incomplete BEE.
[0092] Clause 17. The method of clause 16, wherein each BEE of the plurality of BEEs further comprises one or more triggers, and wherein querying the BEE database to determine the first or the second set of active BEEs comprises, querying the BEE database for one or more triggers that have a condition matching the simulation state update, storing the one or more triggers as a set of active triggers, and querying the BEE database to determine whether any BEEs are active BEEs based on the set of active triggers.
[0093] Clause 18. The method of clause 17, wherein querying the BEE database to determine the

first or the second set of active BEEs further comprises determining one or more inactive triggers in the set of active triggers by comparing a corresponding condition for each active trigger in the set of active triggers to the simulation state update, and removing the one or more inactive triggers from the set of active triggers.
[0094] Clause 19. The method of clause 16, further comprising, when the action represented by the simulation action notification matches the elicited action for the at least one active BEE in the set of active BEEs, determining whether the elicited action has a timing condition, and storing information regarding a timing of the simulation action notification.
[0095] Clause 20. The method of clause 16, further comprising, determining a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met; and for each completed BEE of the set of completed BEEs, storing information regarding the training performance, the information comprising the one or more observable behaviors of the completed BEE.
[0096] Clause 21. A method of evaluating a training performance of a training subject against a competency framework using a simulation, the training performance being performed in a training environment that simulates a physical environment, the method comprising: receiving a simulation state update from a computing system of the training environment;querying a behavioral elicitation event (BEE) database to determine, from a plurality of BEEs stored in the BEE database, a set of active BEEs based upon the simulation state update, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors; receiving a plurality of simulation action notifications from the computing system of the training environment; for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs and storing information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs; determining a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met; for each completed BEE of the set of completed BEEs, storing information regarding the training performance, the information comprising the one or more observable behaviors of the completed BEE; and displaying at least the information regarding the training performance on a user interface.
[0097] Clause 22. The method of clause 21, wherein the user interface comprises a menu to allow selection of a phase of the simulation.
[0098] Clause 23. The method of clause 22, wherein the user interface further displays, for a selected menu item, an observable behavior display for the selected menu item, the observed behavior display including information on elicited actions observed during the simulation.
[0099] Clause 24. The method of clause 23, wherein the observable behaviors display for the selected menu item is expandable.
[0100] Clause 25. The method of clause 24, further comprising displaying information regarding each elicited action for the selected menu item when the observable behaviors display is expanded for the selected menu item.
[0101] Clause 26. The method of clause 25, wherein the information regarding each elicited action comprises a progress bar for each elicited action.
[0102] Clause 27. The method of clause 21, further comprising displaying, on the user interface, summarized information regarding at least the training performance.
[0103] Clause 28. The method of clause 27, wherein the summarized information regarding at least the training performance is displayed upon selection of the selected menu item displayed on the user interface.
[0104] Clause 29. The method of clause 21, wherein the simulation is a flight simulation, and wherein the user interface further comprises a graph showing information on at least one phase of the flight simulation.
[0105] Clause 30. The method of clause 29, wherein the graph illustrates positional information for

a simulated aircraft as a function of time.

[0106] Clause 31. A computing system, comprising: a logic system operably coupled to a training environment that simulates a physical environment; and a computer-readable storage system comprising, a behavioral elicitation event (BEE) database comprising a plurality of BEEs, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors, and instructions executable by the logic system to perform an evaluation of a training performance of a training subject against a competency framework using a simulation, the instructions being executable to receive a simulation state update from a computing system of the training environment, query the BEE database to determine, from the plurality of BEEs, a set of active BEEs based upon the simulation state update, receive a plurality of simulation action notifications, for each simulation action notification, determine whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs, and store information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs, determine a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met, for each completed BEE of the set of completed BEEs, store information regarding training performance, the information regarding the training performance comprising information regarding the completed BEE, and display at least the information regarding the training performance on a user interface.

[0107] Clause 32. The computing system of clause 31, wherein the instructions are further executable to display, on the user interface, a plurality of menu items, each menu item corresponding to a corresponding phase of the simulation.

[0108] Clause 33. The computing system of clause 32, further comprising instructions executable to display, for a selected menu item, an observable behaviors display for the selected menu item, the observable behavior display including information on elicited actions performed by a trainee during the simulation.

[0109] Clause 34. The computing system of clause 33, wherein the observable behaviors display is expandable.

[0110] Clause 35. The computing system of clause 31, wherein the instructions are further executable to display, on the user interface, summarized information regarding the training performance.

[0111] Clause 36. The computing system of clause 31, wherein the simulation is a flight simulation, and wherein the instructions are further executable to display, on the user interface, a graph showing information on at least one phase of the flight simulation, and wherein the graph illustrates positional information for a simulated aircraft as a function of time.

[0112] Clause 37. A method of displaying information related to a flight simulation, the method comprising: displaying on the a user interface an observable behavior display comprising information regarding the one or more observable behaviors of the at least one phase of the flight simulation, the one or more observable behaviors corresponding to a behavioral elicitation event (BEE) that occurs during the flight simulation; and displaying on the user interface information regarding one or more elicited actions performed by a trainee for each observable behavior of the one or more observable behaviors.

[0113] Clause 38. The method of clause 37, wherein the method further comprises displaying on the user interface a menu, the menu comprising selectable menu items each corresponding to a flight phase of the flight simulation.

[0114] Clause 39. The method of clause 37, wherein the observable behavior display for each observable behavior of the one or more observable behaviors is expandable to display an expanded observable behavior display, and wherein the expanded observable behavior display comprises a progress bar for each elicited action displayed.

[0115] Clause 40. The method of clause 37, further comprising displaying on a user interface a

graphical representation of a flight path during at least one phase of the flight simulation.
[0116] It will be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated and/or described may be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. Likewise, the order of the above-described processes may be changed.
[0117] The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. A method of evaluating a training performance of a training subject against a competency framework using a simulation, the training performance being performed in a training environment that simulates a physical environment, the method comprising: receiving a simulation state update from a computing system of the training environment; querying a behavioral elicitation event (BEE) database to determine, from a plurality of BEEs stored in the BEE database, a set of active BEEs based upon the simulation state update, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors; receiving a plurality of simulation action notifications from the computing system of the training environment; for each simulation action notification, determining whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs and storing information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs; determining a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met; for each completed BEE of the set of completed BEEs, storing information regarding the training performance, the information comprising the one or more observable behaviors of the completed BEE; and displaying at least the information regarding the training performance on a user interface.
2. The method of claim 1, wherein the user interface comprises a menu to allow selection of a phase of the simulation.
3. The method of claim 2, wherein the user interface further displays, for a selected menu item, an observable behavior display for the selected menu item, the observed behavior display including information on elicited actions observed during the simulation.
4. The method of claim 3, wherein the observable behaviors display for the selected menu item is expandable.
5. The method of claim 4, further comprising displaying information regarding each elicited action for the selected menu item when the observable behaviors display is expanded for the selected menu item.
6. The method of claim 5, wherein the information regarding each elicited action comprises a progress bar for each elicited action.
7. The method of claim 1, further comprising displaying, on the user interface, summarized information regarding at least the training performance.
8. The method of claim 7, wherein the summarized information regarding at least the training performance is displayed upon selection of the selected menu item displayed on the user interface.
9. The method of claim 1, wherein the simulation is a flight simulation, and wherein the user interface further comprises a graph showing information on at least one phase of the flight simulation.

10. The method of claim 9, wherein the graph illustrates positional information for a simulated aircraft as a function of time.
11. A computing system, comprising: a logic system operably coupled to a training environment that simulates a physical environment; and a computer-readable storage system comprising, a behavioral elicitation event (BEE) database comprising a plurality of BEEs, each BEE of the plurality of BEEs comprising one or more elicited actions and one or more observable behaviors, and instructions executable by the logic system to perform an evaluation of a training performance of a training subject against a competency framework using a simulation, the instructions being executable to receive a simulation state update from a computing system of the training environment, query the BEE database to determine, from the plurality of BEEs, a set of active BEEs based upon the simulation state update, receive a plurality of simulation action notifications, for each simulation action notification, determine whether an action represented by the simulation action notification matches an elicited action for at least one active BEE in the set of active BEEs, and store information regarding the elicited action as a response to the at least one active BEE in the set of active BEEs, determine a set of completed BEEs, the set of completed BEEs comprising each active BEE that has all corresponding elicited actions met, for each completed BEE of the set of completed BEEs, store information regarding training performance, the information regarding the training performance comprising information regarding the completed BEE, and display at least the information regarding the training performance on a user interface.
12. The computing system of claim 11, wherein the instructions are further executable to display, on the user interface, a plurality of menu items, each menu item corresponding to a corresponding phase of the simulation.
13. The computing system of claim 12, further comprising instructions executable to display, for a selected menu item, an observable behaviors display for the selected menu item, the observable behavior display including information on elicited actions performed by a trainee during the simulation.
14. The computing system of claim 13, wherein the observable behaviors display is expandable.
15. The computing system of claim 11, wherein the instructions are further executable to display, on the user interface, summarized information regarding the training performance.
16. The computing system of claim 11, wherein the simulation is a flight simulation, and wherein the instructions are further executable to display, on the user interface, a graph showing information on at least one phase of the flight simulation, and wherein the graph illustrates positional information for a simulated aircraft as a function of time.
17. A method of displaying information related to a flight simulation, the method comprising: displaying on a user interface an observable behavior display comprising information regarding one or more observable behaviors of at least one phase of the flight simulation, the one or more observable behaviors corresponding to a behavioral elicitation event (BEE) that occurs during the flight simulation; and displaying on the user interface information regarding one or more elicited actions performed by a trainee for each observable behavior of the one or more observable behaviors.
18. The method of claim 17, wherein the method further comprises displaying on the user interface a menu, the menu comprising selectable menu items each corresponding to a flight phase of the flight simulation.
19. The method of claim 17, wherein the observable behavior display for each observable behavior of the one or more observable behaviors is expandable to display an expanded observable behavior display, and wherein the expanded observable behavior display comprises a progress bar for each elicited action displayed.
20. The method of claim 17, further comprising displaying on a user interface a graphical representation of a flight path during at least one phase of the flight simulation.